

## MANGALORE INTL, India

WMO: 432840

Lat: 12.961N

Lon: 74.890E

Elev: 103

StdP: 100.10

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 21.0       | 21.8 | 12.2                          | 9.0 | 30.0 | 14.1 | 10.1 | 29.0 | 9.6                   | 26.4 | 8.4 | 27.5 | 2.3                   | 90   | 0.339 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 7.8                    | 34.4            | 24.9 | 34.0 | 25.0 | 33.4 | 24.9 | 27.2                | 31.4 | 26.9 | 31.1 | 26.7 | 30.7 | 4.6                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.2                            | 21.8 | 29.1 | 25.9 | 21.5 | 29.0 | 25.6 | 21.1 | 28.7 | 87.0          | 31.9 | 85.6 | 31.3 | 84.3 | 30.8 | 30.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 7.5           | 6.3  | 5.6 | DB                         | 18.7 | 36.0               | 1.9 | 0.7                                                | 17.4 | 36.6       | 16.3 | 37.0       | 15.2 | 37.4       | 13.9 | 37.9 | (4) |
| (5)               |      |     | WB                         | 15.2 | 28.1               | 0.9 | 0.8                                                | 14.6 | 28.7       | 14.1 | 29.2       | 13.5 | 29.7       | 12.9 | 30.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.6   | 27.2  | 27.9  | 29.0  | 29.7  | 29.2  | 26.7  | 26.0  | 26.1  | 26.7  | 27.3  | 27.9  | 27.4  | (6)  |  |
|                                                                        | DBStd       | 1.56   | 0.99  | 1.03  | 0.94  | 0.96  | 1.30  | 1.52  | 1.10  | 1.01  | 0.93  | 0.94  | 0.98  | 1.00  | (7)  |  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |  |
|                                                                        | CDD10.0     | 6422   | 533   | 502   | 588   | 591   | 596   | 500   | 498   | 501   | 501   | 537   | 536   | 541   | (10) |  |
|                                                                        | CDD18.3     | 3381   | 275   | 269   | 329   | 341   | 337   | 250   | 239   | 242   | 251   | 279   | 286   | 282   | (11) |  |
|                                                                        | CDH23.3     | 33742  | 2524  | 2698  | 3811  | 4332  | 4170  | 2172  | 1838  | 1870  | 2139  | 2606  | 2872  | 2709  | (12) |  |
|                                                                        | CDH26.7     | 12574  | 1046  | 1149  | 1676  | 2054  | 1896  | 576   | 315   | 320   | 508   | 832   | 1099  | 1102  | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.4    | 2.4   | 2.4   | 2.4   | 2.5   | 2.6   | 2.7   | 2.8   | 2.5   | 2.2   | 2.0   | 2.0   | 2.2   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 3730   | 0     | 3     | 4     | 42    | 196   | 947   | 1113  | 807   | 340   | 177   | 74    | 28    | (15) |  |
|                                                                        | PrecMax     | 6651   | 2     | 73    | 38    | 308   | 597   | 2167  | 1719  | 1678  | 938   | 459   | 317   | 281   | (16) |  |
|                                                                        | PrecMin     | 2678   | 0     | 0     | 0     | 0     | 0     | 238   | 608   | 258   | 93    | 32    | 5     | 0     | (17) |  |
|                                                                        | PrecStd     | 926    | 0     | 15    | 9     | 65    | 180   | 386   | 307   | 369   | 213   | 107   | 70    | 59    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 34.2  | 35.3  | 35.0  | 35.2  | 34.7  | 33.0  | 30.4  | 30.5  | 31.7  | 33.1  | 34.0  | 34.1  | (19) |  |
|                                                                        |             | MCWB   | 21.7  | 21.5  | 24.5  | 26.4  | 26.1  | 26.3  | 25.9  | 25.8  | 25.9  | 25.1  | 24.3  | 22.5  | (20) |  |
|                                                                        | 2%          | DB     | 33.1  | 33.9  | 34.0  | 34.3  | 34.0  | 31.7  | 29.8  | 29.7  | 30.7  | 32.0  | 33.0  | 33.1  | (21) |  |
|                                                                        |             | MCWB   | 21.9  | 22.1  | 25.0  | 26.2  | 26.3  | 26.2  | 25.9  | 25.6  | 25.5  | 25.6  | 24.2  | 22.6  | (22) |  |
|                                                                        | 5%          | DB     | 32.3  | 32.9  | 33.3  | 33.9  | 33.3  | 30.6  | 29.1  | 29.1  | 30.0  | 31.2  | 32.3  | 32.3  | (23) |  |
|                                                                        |             | MCWB   | 22.1  | 23.0  | 25.0  | 26.1  | 26.1  | 26.1  | 25.8  | 25.5  | 25.3  | 25.6  | 24.2  | 22.6  | (24) |  |
|                                                                        | 10%         | DB     | 31.4  | 32.1  | 32.8  | 33.2  | 32.8  | 29.7  | 28.5  | 28.5  | 29.3  | 30.5  | 31.6  | 31.7  | (25) |  |
|                                                                        |             | MCWB   | 22.2  | 23.3  | 25.0  | 25.9  | 26.0  | 26.0  | 25.7  | 25.4  | 25.2  | 25.6  | 24.3  | 22.6  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 25.4  | 26.1  | 27.1  | 27.6  | 27.7  | 27.2  | 26.9  | 26.6  | 26.6  | 27.1  | 26.7  | 26.1  | (27) |  |
|                                                                        |             | MCDB   | 29.8  | 31.2  | 32.2  | 33.0  | 32.0  | 30.3  | 28.9  | 28.6  | 29.8  | 30.8  | 30.9  | 30.3  | (28) |  |
|                                                                        | 2%          | WB     | 24.8  | 25.4  | 26.6  | 27.1  | 27.2  | 26.8  | 26.4  | 26.2  | 26.1  | 26.5  | 26.2  | 25.4  | (29) |  |
|                                                                        |             | MCDB   | 29.4  | 30.3  | 31.6  | 32.3  | 31.5  | 29.7  | 28.5  | 28.4  | 29.1  | 30.2  | 30.3  | 29.6  | (30) |  |
|                                                                        | 5%          | WB     | 24.3  | 25.0  | 26.2  | 26.7  | 26.9  | 26.5  | 26.1  | 25.9  | 25.9  | 26.2  | 25.9  | 24.9  | (31) |  |
|                                                                        |             | MCDB   | 28.9  | 29.6  | 31.0  | 31.7  | 31.1  | 29.3  | 28.3  | 28.1  | 28.7  | 29.7  | 29.8  | 29.0  | (32) |  |
|                                                                        | 10%         | WB     | 23.8  | 24.5  | 25.8  | 26.5  | 26.6  | 26.1  | 25.7  | 25.6  | 25.6  | 25.8  | 25.5  | 24.4  | (33) |  |
|                                                                        |             | MCDB   | 28.4  | 29.1  | 30.5  | 31.4  | 30.8  | 28.9  | 27.8  | 27.7  | 28.3  | 29.2  | 29.3  | 28.5  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.2  | 9.7   | 8.6   | 7.8   | 7.1   | 5.0   | 4.3   | 4.4   | 5.5   | 6.8   | 8.5   | 9.8   | (35) |  |
|                                                                        |             | MCDBR  | 10.9  | 10.6  | 8.9   | 8.2   | 7.7   | 6.7   | 5.3   | 5.3   | 6.4   | 7.8   | 9.3   | 10.4  | (36) |  |
|                                                                        |             | MCWBR  | 3.8   | 3.6   | 2.8   | 2.6   | 2.5   | 2.7   | 2.7   | 2.4   | 2.4   | 3.1   | 3.2   | 3.6   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 9.4   | 9.3   | 8.3   | 7.8   | 7.2   | 5.8   | 4.8   | 4.8   | 5.6   | 7.1   | 7.9   | 8.5   | (38) |  |
|                                                                        |             | MCWBR  | 3.7   | 3.4   | 2.7   | 2.9   | 2.8   | 3.0   | 2.8   | 2.6   | 2.6   | 3.2   | 3.1   | 3.1   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.459 | 0.491 | 0.562 | 0.600 | 0.571 | 0.554 | 0.594 | 0.549 | 0.495 | 0.534 | 0.494 | 0.452 | (40) |  |
|                                                                        | taud        |        | 2.213 | 2.126 | 1.963 | 1.897 | 1.978 | 2.034 | 1.913 | 2.042 | 2.216 | 2.079 | 2.173 | 2.263 | (41) |  |
|                                                                        | Ebn at Noon |        | 839   | 827   | 777   | 743   | 752   | 755   | 728   | 772   | 820   | 780   | 800   | 835   | (42) |  |
|                                                                        | Edh at Noon |        | 141   | 159   | 191   | 203   | 184   | 172   | 195   | 173   | 146   | 165   | 145   | 131   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 5.56  | 6.22  | 6.58  | 6.50  | 5.76  | 3.95  | 3.80  | 4.25  | 5.02  | 5.04  | 5.06  | 5.17  | (44) |  |
|                                                                        | RadStd      |        | 0.19  | 0.20  | 0.26  | 0.24  | 0.46  | 0.35  | 0.46  | 0.37  | 0.44  | 0.29  | 0.30  | 0.26  | (45) |  |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page